

A Robust Privacy-Preserving Data Aggregation Scheme for Edge-Supported IIoT

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Abstract—Edge-supported Industrial Internet of Things (IIoT) has received remarkable attention recently since edge computing can not only reduce bandwidth consumption but also decrease the response time of industrial systems. However, the sensed data in the industrial environment is considered private. Thus, the data cannot be directly aggregated at the server due to privacy leakage. Although several privacy-preserving-aggregated schemes have been proposed, their security goals are not strong enough. Besides, most schemes are inefficient for resource-constrained devices. Aiming at solving the abovementioned problems, this article proposes a robust privacy-preserving data aggregation scheme for edge-supported IIoT. Specifically, the scheme adopts the Paillier cryptosystem to protect the privacy of users. Additionally, it utilizes ECDSA signature to support batch verification of multiple signatures from different signers, which significantly improves efficiency.

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Security analysis shows that the proposed scheme not only guarantees the integrity of the data and mutual authentication among entities but also realizes differential privacy protection. Extensive experiments are conducted to compare our scheme with the related work. The results show that our method outperforms most of the compared schemes with respect to communication. Moreover, compared with the related work, our scheme reduces the computational cost by an average of 6.7%, 16.9%, and 27.8% in sensor, edge server, and control center sides, respectively.

Index Terms—Data aggregation, edge computing, Industrial Internet of Things (IIoT), privacy-preserving.

I. INTRODUCTION

THE Internet of Things (IoT) [1] is a system of interrelated devices with certain sensing and computing capabilities, which can collect a large amount of sense data. For the sensed data, cloud computing [2] can provide storage and computation. However, due to the explosion of the data and the limitation of bandwidth, it is difficult for centralized cloud computing to achieve real-time computing. Therefore, edge computing [3] was proposed to preprocess the sensed data at the edge, which evidently reduces bandwidth consumption and response time of the industrial systems.

However, if the sensing data is processed in the edge server in plaintext, the private information of industrial companies will be leaked. For example, the production technology of an industrial company, such as production environment parameters and materials used, etc., is not only commercial secret but also the privacy of the company. Therefore, if the sensed data, such as temperature and pressure, is directly uploaded to edge servers without protection, an adversary can easily obtain the environment parameters by public channel or the compromised edge servers and further infer the production technology. As a result, the company may suffer huge losses. Therefore, the sensed data should be uploaded and processed in a privacy-preserving way. Fortunately, the data aggregation technique can process data in a privacy-preserving way, which not only saves the communication bandwidth but also protects the privacy of sensed data [4] in different environments.

A. Related Work

Recently, a lot of data aggregation schemes in smart grid [5], [6], [7], [8], [9], [10], [11] have been proposed for various

functionalities and security features. In 2018, Abdallah et al. [5] proposed a lightweight lattice-based data aggregation scheme. This scheme migrates encryption and aggregation operations to smart household appliances (APs), which significantly improves security features. In 2019, Saleem et al. [6] designed a fog-enabled scheme for a smart grid IoT network utilizing the Paillier homomorphic algorithm. The scheme not only supports the verification of received messages but also reduces communication overhead. However, it is vulnerable to differential attacks. In order to resolve this issue, Li et al. [7] presented a novel secure and efficient IoT data aggregation scheme, which adopts a replacement circuit based on oblivious transmission to prevent the server from knowing which device has obtained the generated secret and avoid the collusion between the client and the server. However, this scheme cannot resist denial-of-service (DoS) attacks. Chen et al. [8] proposed a multidimensional data aggregation scheme for the smart grid. It allows the server to aggregate multiple types of data, which is not only more in line with real-world applications but also can obtain more valuable information. However, the scheme adopts bilinear pairing, which causes a huge computational overhead for smart meters. Liu et al. [9] presented a privacy-preserving aggregation scheme with function query for fog-computing-based smart grid, which provides users or the service provider with function queries on encrypted data. However, this scheme has a large communication and computation overhead.

In order to achieve efficient data aggregation in the smart grid, Su et al. [10] designed a lightweight and communication-efficient data aggregation scheme. In order to improve the computation efficiency, the authors adopted the widely-used masking-value-based method to protect the privacy and confidentiality of the sensed data in the smart grid. Even though the scheme remarkably improves efficiency, the computational cost of the registration is still heavy and the data aggregation results cannot be verified. In 2022, Zhang et al. [11] proposed a verifiable privacy-preserving multitype data aggregation in the smart grid, they modified the Paillier cryptosystem to support multiple types of data aggregation and meanwhile altered the BLS signatures to achieve the verifiability of the aggregated results in the aggregator.

Even though fog/edge computing is a promising solution to improve the service of mobile providers, the security of the data in the edge nodes remains an important concern [12]. The data generated by smart grid-related applications are very sensitive [13], and the privacy, confidentiality, and integrity of these data should be strictly maintained. Due to the natural features of decentralization, immutability, and transparency, blockchain has been integrated with edge computing for IoT. The authors in [14] and [15] have proposed a high-level architecture by combining blockchain and edge computing for IoT, which can significantly improve the security of various applications by introducing more security features, such as access authentication, data privacy preservation, attack detection, and trust management.

Later, some researchers designed data aggregation schemes [16], [17], [18], [19] in smart grid with high-level security by combining blockchain and edge computing. In order to meet the requirement of the microgrid, Luo

et al. [16] proposed a secure data aggregation scheme based on homomorphic encryption and the practical Byzantine fault tolerance (PBFT) consensus by leveraging blockchain. The scheme securely implements data aggregation and power dispatching in microgrids without any trusted third party. Similarly, in order to get rid of the centralized model with a trusted third party, Regueiro et al. [17] designed a privacy-enhanced data aggregation protocol based on blockchain and homomorphic encryption. In [18], Lu et al. devised an edge-blockchain-assisted data aggregation scheme that combines edge computing and blockchain to ensure data security and privacy protection at the edge layer. Then, Chen et al. [19] proposed a double-blockchain-assisted secure and anonymous data aggregation scheme. The scheme adopted a three-tier architecture that integrates fog computing and the blockchain, which achieves fine-grained and two-level data aggregation. Moreover, robustness has been considered in various types of schemes [20], [21], [22] as a high level of security guarantee. Therefore, robustness should be considered more in data aggregation schemes to achieve more security features. The related work is summarized in Table I.

Despite the existing data aggregation schemes, there are some limitations to be solved, which are as follows.

- 1) Most schemes do not support the efficient authentication of both data integrity and device identity through batch signature verification.
- 2) Most schemes cannot provide security guarantees of privacy protection, data integrity, identity authentication, differential privacy, and resistance to replay attacks simultaneously.
- 3) The existing schemes suffer inefficient communication and computation as the time-consuming bilinear pairing is often used.

At the edge of the network, especially in edge-supported Industrial IoT (IIoT), terminal devices, e.g., smart meters, mainly face various security threats, e.g., data privacy leakage, and suffer inefficient communication and computation where they usually have poor resources. However, most existing work cannot achieve sufficiently efficient communication and computation at the terminal device side at the base of meeting the necessary security features. To meet the scenario requirements, we always consider balancing the security guarantees and the efficiency of the devices. Regarding the security features, we have covered the most fundamental ones to make the scheme robust sufficiently, including privacy, integrity, authentication, resistance to replay attacks, and differential privacy. As for efficiency, we adopt a batch signature verification algorithm for the elliptic curve digital signature algorithm (ECDSA) to speed up authentication. Meanwhile, we manage to avoid time-consuming operations, e.g., bilinear pairing, and perform precomputations as possible. As a result, the efficiency is significantly improved.

B. Our Contributions

To address the above problems, we propose a robust privacy-preserving data aggregation scheme for edge-supported IIoT, called PPDA, in this work. The Paillier cryptosystem and elliptic curve digital signature algorithm are used for privacy protection

TABLE I
SUMMARY OF THE RELATED WORK

Schemes	Architecture	Main techniques	Non-repudiation	IdenAuth	DP	Functionality
[5]	EoT	Lattice-based HE	✓	✓	×	Sum
[6]		HE, MAC	×	✓	×	Sum
[7]		HE, Oblivious transmission	×	×	×	Sum
[8]		HE	×	×	×	Sum, Variance
[9]		Double trapdoor Cryptosystem	✓	✓	✓	Sum, Function query
[10]		TC secret sharing	×	×	×	Sum
[11]		HE, Modified BLS signature	✓	✓	×	Sum, Variance
[16]	BEoT	HE, PBFT	×	×	×	Sum, Power dispatching
[17]		HE, Smart Contracts	×	×	×	Sum
[18]		HE, Hash Chain	×	×	×	Sum
[19]		HE, BaAggSig	✓	✓	×	Sum

In the first row, IdenAuth and DP denote identity authentication and differential privacy, respectively. In the other part, HE, MAC, PBFT, BaAggSig, and TC secret sharing represent homomorphic encryption, message authentication code, practical byzantine fault tolerance consensus, batch aggregation signature, and threshold changeable secret sharing, respectively.

and authentication, respectively. Moreover, edge computing is also adopted to reduce the computation and communication costs of the terminal device. Our contributions can be summarized as the following.

- 1) Based on the Paillier cryptosystem and the modified ECDSA signature, we design a robust privacy-preserving data aggregation scheme for IIoT (PPDA). More importantly, the scheme enables efficient authentication for data integrity and device identity via batch signature verification.
- 2) With respect to security features, we not only achieve privacy protection, data integrity, identity authentication, and resistance against replay attacks but also protect the sensed data from differential attacks, ensuring the high-security level of the sensed data.
- 3) We efficiently implement the PPDA and compare the performance of PPDA with that of other related work. The experiment results show that PPDA is efficient in terms of computation and communication.

C. Organization of the Article

The rest content of this article is organized as follows. Section II presents the preliminaries. The system model, security model, and design goals are illustrated in Section III. Section IV demonstrates the specific details of the proposed scheme PPDA. The correctness proof, security analysis, and security comparison are given in Section V. Section VI compares the performance of our scheme with that of the other six related schemes. Finally, Section VII concludes this article.

II. PRELIMINARIES

In this section, we briefly introduce the Paillier cryptosystem [23], differential privacy [24], and signature algorithm with batch verification [25], which are used as the basis of the proposed scheme.

A. Paillier Cryptosystem

The Paillier cryptosystem [23] is widely used in privacy protection due to its homomorphic properties and resistance to chosen-plaintext attacks. It consists of three algorithms as follows.

- 1) *Key Generation*: Given a security parameter k_1 , two large prime numbers p_0 and q_0 are chosen satisfying $|q_0| = |p_0| = k_1$. Let $N = p_0 \cdot q_0$ and $\lambda = \text{lcm}(p_0 - 1, q_0 - 1)$. Then, $L(x)$ is defined as $L(x) = \frac{x-1}{N}$, and $\mu = (L(g^\lambda \bmod N^2))^{-1}$ is calculated, where g is a generator of $Z_{N^2}^*$. The public key and private key pairs are (N, g) and (λ, μ) , respectively.
- 2) *Encryption*: The ciphertext of a message $M \in Z_N$ can be calculated as $C = g^M \cdot r^N \bmod N^2$, where $r \in Z_N^*$.
- 3) *Decryption*: The underlying plaintext M of the ciphertext C can be recovered as $M = L(C^\lambda \bmod N^2) \cdot \mu \bmod N$.

B. Differential Privacy

The differential privacy technique [24] is first proposed in 2006. It can protect the privacy of users on neighboring datasets by adding appropriate noise.

Definition 1. ϵ -differential privacy: A randomized algorithm K gives ϵ -difference privacy, in any dataset D_1 and D_2 that differs by at most one element, for any output $t \in \text{Range}(K)$, satisfying the following formula:

$$\Pr[K(D_1) = t] \leq \exp(\epsilon) \cdot \Pr[K(D_2) = t].$$

Definition 2. Global sensitivity: For a query function $f : D \rightarrow R$, the global sensitivity of the query function f on the neighboring datasets D_1 and D_2 is

$$\Delta f = \max |f(D_1) - f(D_2)|.$$

The Laplace mechanism satisfying ϵ -differential privacy is denoted as $\text{Lap}(\frac{\Delta f}{\epsilon})$. Generally speaking, the smaller the value of the privacy budget ϵ is, the stronger the privacy protection provided by the mechanism is.

Because of the above properties, a scheme satisfying differential privacy can resist the differential attack.

C. Signature Algorithm With Batch Verification

The signature algorithm with batch verification supports the verification of multiple signatures from different signers at once, which significantly improves the efficiency of verification compared with general signatures. In our scheme, we utilize the modified ECDSA [25] to support batch verification for signatures from different signers. The details of the algorithm are described as follows.

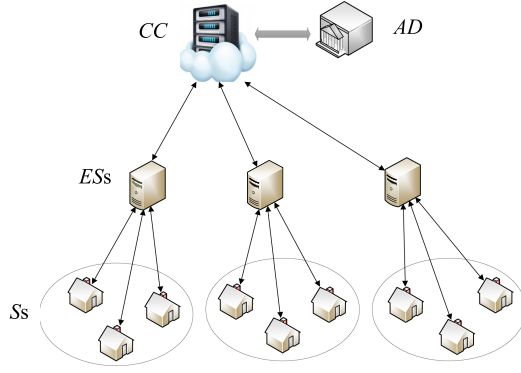


Fig. 1. System model of the proposed scheme.

- 1) $((Q_i, d_i)_{i \in [1, t]}, G_q(a, b)) \leftarrow \text{ParaGen}(k_2, k_3)$: A signer U first selects a large prime number q of size k_2 and random numbers $a, b \in \mathbb{Z}_q$ as the coefficients of equation $y^2 = x^3 + ax + b \pmod q$ to initialize the signature algorithm. Group $G_q(a, b)$ is defined in points of the equation with its order n of k_3 bit. $G = (x_g, y_g)$ is chosen as the generator of $G_q(a, b)$ and O is the point at infinity in $G_q(a, b)$. Suppose there are t signers. Each signer U_i selects a random number $d_i \in [1, n - 1]$ as its private key and further generates its public key $Q_i = d_i G$.
- 2) $(P_i, s_i) \leftarrow \text{Sig}(d_i, m_i)$: To sign a message m_i , a signer U_i selects a random number $k_i \in [1, n - 1]$ to compute $P_i = (x_i, y_i) = k_i G$ and $r_i = x_i \pmod n$ satisfying $r_i \neq 0$. Next, U_i computes the hash value of m_i as $e_i = H(m_i)$, where $H(\cdot)$ represents the SHA-256 algorithm. Afterward, s_i is computed as $s_i = k_i^{-1}(e_i + d_i r_i) \pmod n$, satisfying $s_i \neq O$. Finally, U_i outputs (m_i, P_i, s_i) as the signature of the message m .
- 3) $(\text{Accept/Reject}) \leftarrow \text{Ver}((m_i, P_i, s_i), Q_i)_{i \in [1, t]}$: Given signatures (m_i, P_i, s_i) , $i \in [1, t]$, the verifier V first checks if r_i and $s_i \in [1, n - 1]$, and then computes the hash value of message m_i as $e_i = H(m_i)$. Finally, V checks if the equation $(\sum_{i=1}^t u_{1i})G + \sum_{i=1}^t u_{2i}Q_i = \sum_{i=1}^t P_i$ holds, where $u_{1i} = e_i s_i^{-1}$ and $u_{2i} = r_i s_i^{-1}$, $i \in [1, n - 1]$. If so, V accepts these signatures. Otherwise, V rejects these signatures since at least one of the signatures is invalid.

III. SYSTEM AND SECURITY MODELS AND DESIGN GOALS

In this section, we introduce the system model and security model of our scheme. Then, we illustrate the design goals of our scheme.

A. System Model

Fig. 1 shows the system structure of edge-supported IIoT, and there are four types of entities involved in this system, i.e., an administrator (AD), a cloud center (CC), edge servers (ES)s, and sensors (Ss).

- 1) *Administration (AD)*: The AD is the initiator and final processor of the task in the entire system. It analyzes the collected sensed data on the cloud server and provides corresponding decisions.

- 2) *Cloud center (CC)*: The CC has sufficient computing and communication resources. After decrypting the aggregated ciphertext, CC processes the data according to the requirements of AD.
- 3) *Edge servers (ESs)*: ESs act as aggregators in IIoT, which bridge CC and sensors Ss. They aggregate data from Ss in a secure manner and upload the aggregated data ciphertext to CC.
- 4) *Sensors (Ss)*: Ss usually have limited computing and communication resources. They collect data from the industrial environment and encrypt the sensed data. Then, they transmit the encrypted data to the corresponding ESs.

B. Security Model

In our model, AD is considered fully trusted. CC, ESs, and Ss are all assumed as semi-honest entities, i.e., each entity performs the protocol specification but may attempt to infer sensors Ss' private information. The security goals of this scheme are summarized as follows.

- 1) *Privacy protection*: As the important privacy of an industrial company, the sensed data should not be obtained by other entities, i.e., the original data cannot be recovered by other entities except the sensors where the sensed data are generated.
- 2) *Identity authentication*: In our model, all ESs and Ss are not fully trusted. Therefore, each of them should first register in the system and, then, must be authenticated their legitimate identity by their communication partners.
- 3) *Data integrity*: Messages can be easily modified by malicious adversaries. As a result, any modifications to the message should be checked by the message receiver to ensure data integrity.
- 4) *Differential privacy*: The adversary may obtain the data of a single S through the differential attack during the transmission of the sensed data. Therefore, our scheme should resist the differential attack by adopting differential privacy technology.
- 5) *Resistance to replay attacks*: The adversary may attack the system or recover privacy by performing replay attacks. So, our scheme should be resistant to replay attacks.

C. Design Goals

In order to support privacy-preserving data aggregation, our PPDA should achieve the following goals.

- 1) *Security and privacy*: The electricity consumption data is the privacy information of users and provides valuable power supply information for power dispatching. As a result, the proposed scheme should achieve the security goals mentioned earlier.
- 2) *Aggregation correctness*: In the proposed scheme, ES should correctly aggregate the encrypted user data as well as the corresponding signatures. Finally, CC can get the correct aggregated results.
- 3) *High performance*: The smart meters/sensors are typically computing and communication resource-constrained devices. Therefore, the proposed scheme should achieve efficient communication and computation at a practical level, especially for smart meters/sensors.

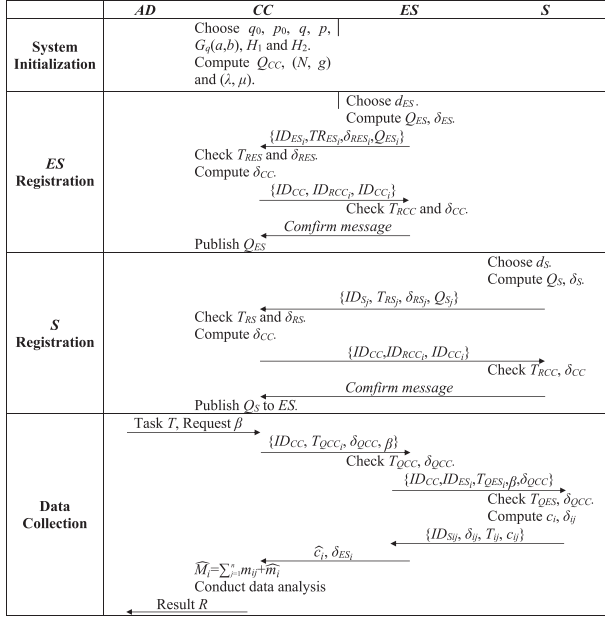


Fig. 2. Workflow of PPDA.

TABLE II
NOTATIONS

Notation	Descriptions
k_1, k_2	Two security parameters
p_0, q_0, p, q	Prime numbers
n	Order of the group $G_q(a, b)$
a, b	Coefficients of elliptic curve
N, g	Public key of Paillier cryptosystem
λ, μ	Private key of Paillier cryptosystem
G	Cycle group with order q
$d_{CC}, d_{ES_i}, d_{S_j}$	Signature private key of CC, ES_i, S_j
$Q_{CC}, Q_{ES_i}, Q_{S_j}$	Signature public key of CC, ES_i, S_j
H_1, H_2	One-way hash functions
ES	Edge server
CC	Cloud center
S	Sensor
$ID_{CC}, ID_{ES_i}, ID_{S_j}$	Identifications of CC, ES_i, S_j
TR_{CC_i}, TR_{CC_j}	Registration timestamps of CC
TR_{ES_i}, TR_{S_j}	Registration timestamps of ES_i, S_j
$\delta_{CC_i}, \delta_{CC_j}$	Registration signatures of CC
$\delta_{RES_i}, \delta_{S_j}$	Registration signatures of ES_i, S_j
β	Collection request
T_{QCC}, T_{QES_i}	Timestamps of CC, ES_i
$\delta_{QCC}, \delta_{QES_i}$	Signatures of CC, ES_i
δ_{ij}	S_{ij} 's signature in data generation
δ_{ES_i}	ES_i 's signature in data aggregation
$\bar{M}_i$	Differential data recovered by CC

IV. PROPOSED SCHEME

In this section, we introduce our privacy-preserving data aggregation scheme for edge-supported IIoT in detail, and the workflow of PPDA is also shown in Fig. 2. Additionally, the notations used in the scheme and the corresponding meanings are listed in Table II.

A. System Initialization

According to the key generation algorithm of the Paillier cryptosystem in Section II-A, CC chooses two prime numbers p_0, q_0 to generate the public key pair (N, g) and the corresponding

private key pair (λ, μ) , where $|p_0| = |q_0| = k_1$. According to the modified ECDSA signature in Section II-C, CC chooses a prime number q and an elliptic curve $G_q(a, b)$ defined by $y^2 = x^3 + ax + b \mod q$, where $a, b \in Z_q$ and $4a^3 + 27b^2 \neq 0$. Next, a generator G of order n of the group $G_q(a, b)$ is chosen. Then, CC selects a private key $d_{CC} \in [1, n-1]$ and further computes the corresponding public key $Q_{CC} = d_{CC}G$. After that, a prime number p and two hash functions: $H_1 : \{0, 1\}^* \rightarrow Z_p^*$, $H_2 : \{0, 1\}^* \rightarrow \{0, 1\}^*$ are selected, where $|p| = k_2$. Finally, the parameters $\{N, g, H_1, H_2, q, a, b, G, n\}$ are published to the whole system.

B. Registration

In this section, we describe the detailed registration process of ES s and S s.

1) **Registration of ES s:** To register to the system, $ES_i, i \in [1, l]$, first chooses a private key $d_{ES_i} \in [1, n-1]$ and calculates the corresponding public key $Q_{ES_i} = d_{ES_i}G$. Next, it generates a signature $\delta_{RES_i} = (P_{RES_i}, S_{RES_i}) = \text{sig}(d_{ES_i}, H_1(ID_{ES_i} || TR_{RES_i}))$ based on its private key d_{ES_i} , identity ID_{ES_i} , and the current timestamp TR_{RES_i} , where $\text{sig}(\cdot, \cdot)$ represents the signature algorithm with batch verification in Section II-C. Then, ES_i sends the message $\{ID_{ES_i}, TR_{RES_i}, \delta_{RES_i}, Q_{ES_i}\}$ to CC . If the timestamp TR_{RES_i} is fresh and the signature δ_{RES_i} is valid, CC computes a signature $\delta_{CC_i} = (P_{CC_i}, S_{CC_i}) = \text{sig}(d_{CC}, H_1(ID_{CC} || TR_{RCC_i}))$ with its private key d_{CC} , identity information ID_{CC} , and timestamp TR_{RCC_i} . Finally, CC returns the message $\{ID_{CC}, TR_{RCC_i}, \delta_{CC_i}\}$ to ES_i . If the signature δ_{CC_i} is valid and the timestamp TR_{RCC_i} is fresh, ES_i returns a confirmation message to CC . On receiving the confirmation message from ES_i , CC publishes the public key Q_{ES_i} of ES_i . Until now, ES_i has completed the registration.

2) **Registration of S s:** The registration of S s is similar to that of ES s. In the registration, $S_j, j \in [1, t]$, chooses its private key $d_{S_j} \in [1, n-1]$, and calculates the corresponding public key $Q_{S_j} = d_{S_j}G$. Then, S_j calculates a signature $\delta_{RS_j} = (P_{RS_j}, S_{RS_j}) = \text{sig}(d_{S_j}, H_1(ID_{S_j} || TR_{RS_j}))$, where d_{S_j}, ID_{S_j} , and TR_{RS_j} are its private key, identity information, and the current timestamp, respectively. Then, S_j sends the message $\{ID_{S_j}, TR_{RS_j}, \delta_{RS_j}, Q_{S_j}\}$ to CC . When receiving the message, CC checks the timestamp TR_{RS_j} and the signature δ_{RS_j} . If the timestamp is fresh and the signature is valid, CC accepts the public key of S_j and calculates a signature $\delta_{CC_j} = (P_{CC_j}, S_{CC_j}) = \text{sig}(d_{CC}, H_1(ID_{CC} || TR_{RCC_j}))$ with its private key d_{CC} , identity ID_{CC} , and timestamp TR_{RCC_j} . Finally, CC returns the message $\{ID_{CC}, TR_{RCC_j}, \delta_{CC_j}\}$ to S_j . If the timestamp TR_{RCC_j} is legal and the signature δ_{CC_j} is valid, S_j returns a confirmation message to CC , and CC publishes the public key Q_{S_j} of S_j to the corresponding ES . Now, S_j has completed the registration.

C. Data Collection

1) **Data Request:** AD first sends data analysis task T to CC . Simultaneously, it also sends a data collection request β to ES s and S s through CC . We suppose that 1

CC , l ES s, and t S s are involved in the data collection phase. CC calculates a signature $\delta_{QCC} = (P_{CC}, s_{CC}) = \text{sig}(d_{CC}, H_1(\text{ID}_{CC} \| T_{QCC} \| \beta))$ according to its private key d_{CC} , the collection request β , its identity ID_{CC} and timestamp T_{QCC} . Then, it broadcasts the message $\{\text{ID}_{CC}, T_{QCC}, \delta_{QCC}, \beta\}$ to all ES s. Each ES_i accepts the request if both the timestamp T_{QCC} and the signature δ_{QCC} are valid. After that, ES_i generates a signature $\delta_{QES_i} = (P_{QES_i}, s_{QES_i}) = \text{sig}(d_{ES_i}, H_1(\beta \| \text{ID}_{CC} \| \text{ID}_{ES_i} \| T_{QES_i}))$ with its private key d_{ES_i} , the collection request β , identity ID_{ES_i} , and ID_{CC} and timestamp T_{QES_i} . Finally, ES_i broadcasts the message $\{\text{ID}_{CC}, \text{ID}_{ES_i}, T_{QES_i}, \beta, \delta_{QES_i}\}$ to its subordinated S s. If the timestamp T_{QES_i} and signature δ_{QES_i} are both valid, S s accept the message and begin to report data.

2) Data Generation and reporting: For the collected data m_{ij} , S_{ij} (the j th sensors of the i th edge server, $i \in \{1, 2, \dots, l\}$, $j \in \{1, 2, \dots, t\}$) chooses a random number $R_{ij} \in Z_N^*$ and encrypts the data as $c_{ij} = g^{m_{ij}} \cdot R_{ij}^N \bmod N^2$. Besides, S_{ij} generates a signature $\delta_{ij} = (P_{ij}, s_{ij}) = \text{sig}(d_{S_{ij}}, H_1(\text{ID}_{S_{ij}} \| c_{ij} \| T_{ij}))$ according to the signature algorithm with batch verification described in Section II-C. Finally, S_{ij} sends the message $\{\text{ID}_{S_{ij}}, \delta_{ij}, T_{ij}, c_{ij}\}$ to ES_i .

3) Data Aggregation: Upon receiving t messages from its subordinated S s, ES_i accepts messages if timestamps are fresh and the algorithm $\text{Ver}(\cdot, \cdot)$, on the input of all signatures $\delta_{ij} = (P_{ij}, s_{ij})$ and all the corresponding public keys Q_{ij} , outputs Accept, i.e., ES_i verifies all the signatures by checking if the equation holds, i.e., $\sum_{j=1}^t H(c_{ij}) s_{ij}^{-1} G + \sum_{j=1}^t r_{ij} s_{ij}^{-1} Q_{S_{ij}} = \sum_{j=1}^t P_{ij}$, where r_{ij} is P_{ij} 's x -coordinate modulo n . Then, ES_i performs an aggregation operation on t ciphertexts contained in t messages and gets the aggregated ciphertext $c_i = g^{\sum_{j=1}^t m_{ij}} \cdot (\prod_{j=1}^t R_{ij})^N$. In order to achieve differential privacy, ES_i adds Laplace distributed noise $\hat{m}_i$ into c_i to generate a new aggregated ciphertext $\hat{c}_i = c_i \cdot g^{\hat{m}_i} = g^{\sum_{j=1}^t m_{ij} + \hat{m}_i} \cdot (\prod_{j=1}^t R_{ij})^N$.

After that, ES_i further calculates the signature $\delta_{ES_i} = (P_i, s_i) = \text{sig}(d_{ES_i}, H_1(\text{ID}_{ES_i} \| \hat{c}_i \| T_{ES_i}))$, where ID_{ES_i} is the identity and T_{ES_i} is the current timestamp. Finally, ES_i sends $\{\text{ID}_{ES_i}, \delta_{ES_i}, T_{ES_i}, \hat{c}_i\}$ to CC .

4) Data Decryption and Response: Upon receiving all messages from l ES s, CC accepts messages if the timestamps are fresh and the algorithm $\text{Ver}(\cdot, \cdot)$, on the input of all signatures $\delta_{ES_i} = (P_i, s_i)$ and the corresponding public keys Q_{ES_i} , outputs Accept, i.e., CC checks if $\sum_{j=1}^t H(\hat{c}_i) s_i^{-1} G + \sum_{j=1}^t r_i s_i^{-1} Q_{ES_i} = \sum_{j=1}^t P_i$ holds, where r_i is P_i 's x -coordinate modulo n . If messages are valid, CC further recovers the aggregated data $\hat{M}_1, \hat{M}_2, \dots, \hat{M}_l$ by decrypting the ciphertexts $\hat{c}_1, \hat{c}_2, \dots, \hat{c}_l$ with the private key (λ, μ) as $\hat{M}_i = L(\hat{c}_i^\lambda \bmod N^2) \cdot \mu \bmod N = \sum_{j=1}^n m_{ij} + \hat{m}_i, i \in [1, l]$.

After obtaining the aggregated data, CC performs the data analysis task T receiving from AD by analyzing the aggregated data. Finally, CC gets the data analysis result R of task T and further sends R to AD .

V. PROOF AND ANALYSIS

In this section, we prove the correctness of key equations in our scheme, analyze the security features that our PPDA

provides, and compare the security features of our PPDA and other related schemes. The details are elaborated as follows.

A. Correctness Proof

We mainly give the correctness analysis of batch verification of signatures and the decryption process of our scheme.

1) Batch Verification on ES Side: For messages $\{\text{ID}_{S_{ij}}, \delta_{ij}, T_{ij}, c_{ij}\}$ from S_{ij} , $j \in [1, t]$, where $\delta_{ij} = (P_{ij}, s_{ij}) = \text{sig}(d_{S_{ij}}, H_1(\text{ID}_{S_{ij}} \| c_{ij} \| T_{ij}))$, $P_{ij} = k_{ij}G = (x_{ij}, y_{ij})$, $s_{ij} = k_{ij}^{-1}(H(c_{ij}) + d_{S_{ij}}r_{ij}) \bmod n$ and $r_{ij} = x_{ij} \bmod n$, ES_i can verify

$$\begin{aligned} & \left(\sum_{j=1}^t H(c_{ij}) s_{ij}^{-1} \right) G + \sum_{j=1}^t r_{ij} s_{ij}^{-1} Q_{S_{ij}} \\ &= \left(\sum_{j=1}^t H(c_{ij}) s_{ij}^{-1} \right) G + \sum_{j=1}^t r_{ij} s_{ij}^{-1} d_{S_{ij}} G \\ &= \left(\sum_{j=1}^t H(c_{ij}) s_{ij}^{-1} + \sum_{j=1}^t r_{ij} s_{ij}^{-1} d_{S_{ij}} \right) G \\ &= \sum_{j=1}^t (H(c_{ij}) s_{ij}^{-1} + r_{ij} s_{ij}^{-1} d_{S_{ij}}) G \\ &= \sum_{j=1}^t s_{ij}^{-1} (H(c_{ij}) + r_{ij} d_{S_{ij}}) G \\ &= \sum_{j=1}^t k_{ij} G = \sum_{j=1}^t P_{ij}. \end{aligned}$$

Therefore, the batch verification of signatures on the ES side is correct.

2) Batch Verification on CC Side: For messages $\{\text{ID}_{ES_i}, \delta_{ES_i}, T_{ES_i}, \hat{c}_i\}$ from ES_i , $i \in [1, l]$, where $\delta_{ES_i} = (P_i, s_i) = \text{sig}(d_{ES_i}, H_1(\text{ID}_{ES_i} \| \hat{c}_i \| T_{ES_i}))$, $P_i = k_i G = (x_i, y_i)$, $s_i = k_i^{-1}(H(\hat{c}_i) + d_{ES_i}r_i) \bmod n$ and $r_i = x_i \bmod n$, CC can verify

$$\begin{aligned} & \left(\sum_{i=1}^l H(\hat{c}_i) s_i^{-1} \right) G + \sum_{j=1}^t r_i s_i^{-1} Q_{ES_i} \\ &= \left(\sum_{i=1}^l H(\hat{c}_i) s_i^{-1} \right) G + \sum_{j=1}^t r_i s_i^{-1} d_{ES_i} G \\ &= \left(\sum_{i=1}^l H(\hat{c}_i) s_i^{-1} + \sum_{j=1}^t r_i s_i^{-1} d_{ES_i} \right) G \\ &= \sum_{i=1}^l (H(\hat{c}_i) s_i^{-1} + r_i s_i^{-1} d_{ES_i}) G \\ &= \sum_{i=1}^l s_i^{-1} (H(\hat{c}_i) + r_i d_{ES_i}) G \\ &= \sum_{i=1}^l k_i G = \sum_{i=1}^l P_i. \end{aligned}$$

Therefore, the batch verification of signatures on the CC side is correct.

3) Correctness of Decryption: According to the work in [23], assuming that x is a nonnegative integer, we can get $(1 + N)^x = 1 + xN \bmod N^2$. Therefore, we have

$$\begin{aligned}
 \hat{M}_i &= L(\hat{c}_i^\lambda \bmod N^2) \cdot \mu \bmod N \\
 &= L(\hat{c}_i^\lambda \bmod N^2) \cdot (L(g^\lambda \bmod N^2))^{-1} \bmod N \\
 &= \frac{L(\hat{c}_i^\lambda \bmod N^2)}{L(g^\lambda \bmod N^2)} \bmod N \\
 &= \frac{L\left(\left(g^{\sum_{j=1}^n m_{ij} + \hat{m}_i} \cdot \left(\prod_{j=1}^t R_{ij}\right)^N\right)^\lambda \bmod N^2\right)}{L(g^\lambda \bmod N^2)} \bmod N \\
 &= \frac{L\left(g^{\lambda(\sum_{j=1}^n m_{ij} + \hat{m}_i)} \cdot \left(\prod_{j=1}^t R_{ij}\right)^{\lambda N} \bmod N^2\right)}{L(g^\lambda \bmod N^2)} \bmod N \\
 &= \frac{L\left(g^{\lambda(\sum_{j=1}^n m_{ij} + \hat{m}_i)} \cdot \left(\prod_{j=1}^t R_{ij}\right)^{\lambda N} \bmod N^2\right)}{L(g^\lambda \bmod N^2)} \bmod N \\
 &= \frac{L\left(g^{\lambda(\sum_{j=1}^n m_{ij} + \hat{m}_i)} \bmod N^2\right)}{L(g^\lambda \bmod N^2)} \bmod N \\
 &= \frac{1 + \lambda\alpha N (\sum_{j=1}^n m_{ij} + \hat{m}_i) - 1}{1 + \lambda\alpha N - 1} \bmod N \\
 &= \sum_{j=1}^n m_{ij} + \hat{m}_i.
 \end{aligned}$$

Therefore, the decryption process and the corresponding decryption results on the CC side are correct.

B. Security Analysis

1) Privacy Protection: In our scheme, S_{ij} encrypts its sensed data m_{ij} to get the ciphertext c_{ij} and then sends the ciphertext to ES_i . ES_i cannot recover the original data m_{ij} from the Paillier ciphertext c_{ij} without the private key as the Paillier cryptosystem is secure against the chosen plaintext attack. Therefore, the privacy of S_{ij} , i.e., the private data m_{ij} , can be protected on the ES_i side.

After all ES s perform the aggregation operation, CC collects all the aggregated ciphertexts and decrypts them to get the aggregated data $\hat{m}_i = L(c_i^\lambda \bmod N^2)\mu \bmod N = \sum_{j=1}^n m_{ij} + \hat{m}_i$. Since the aggregation operation has been performed on the ES_i side, CC can only obtain the data sum of all subordinated S s of ES_i . From the data sum, CC cannot recover the collected data m_{ij} of any single sensor S_{ij} .

Based on the above analysis, we can draw the conclusion that PPDA achieves privacy protection for S s' data.

2) Authentication and Unforgeability: In the PPDA, authentication and unforgeability can be guaranteed by the digital signature technique.

First, ES_i and S_{ij} submit $\{ID_{ES_i}, T_{RES_i}, \delta_{RES_i}, Q_{ES_i}\}$ and $\{ID_{S_{ij}}, T_{RS_{ij}}, \delta_{RS_{ij}}, Q_{S_{ij}}\}$ to CC for registration, respectively. Once their signatures δ_{RES_i} and $\delta_{RS_{ij}}$ are verified successfully, CC accepts their registration. In the data collection, upon

receiving the request $\{ID_{CC}, T_{QCC}, \delta_{QCC}, \beta\}$ from CC , ES_i can verify the signature δ_{QCC} by using the public key of CC . Since $\delta_{QCC} = \text{sig}(d_{CC}, H_1(ID_{CC} \| T_{QCC} \| \beta))$, any adversary cannot forge a legal data collection request without knowing the private key d_{CC} of CC . Similarly, S_{ij} can also verify the validity of the message sender's identity when it receives $\{ID_{CC}, ID_{ES_i}, T_{QES_i}, \beta, \delta_{QES_i}\}$ from ES_i .

In the data generation and report, S_{ij} encrypts the plaintext m_{ij} to c_{ij} , generates the signature δ_{ij} by using the private key $d_{S_{ij}}$ with current timestamp T_{ij} , and sends the message $\{ID_{S_{ij}}, \delta_{ij}, T_{ij}, c_{ij}\}$ to S_{ij} . When receiving messages, ES_i can verify all the signatures of the received messages at once. The processes of the verification can be seen in Section V-A1.

Similarly, the validity of messages $\{ID_{ES_i}, ES_i, \delta_{ES_i}, \hat{c}_i\}$, $i \in [1, l]$, can be authenticated by the CC via batch verification.

3) Data Integrity: During the data generation and report phase, S_{ij} sends the message $\{ID_{S_{ij}}, \delta_{ij}, T_{ij}, c_{ij}\}$ to ES_i , where $\delta_{ij} = H_1(d_{S_{ij}}, ID_{S_{ij}} \| c_{ij} \| T_{ij})$. Without the private key $d_{S_{ij}}$, the adversary cannot forge a valid signature. As a result, any modification on c_{ij} can be checked by verifying the signature. Therefore, the integrity of data transmitted between the S side and the ES side can be guaranteed by the batch verification mechanism of signatures.

Similarly, in the data aggregation phase, the ES_i sends the message $\{ID_{ES_i}, \delta_{ES_i}, T_{ES_i}, \hat{c}_i\}$ to CC , where $\delta_{ES_i} = \text{sig}(d_{ES_i}, H_1(ID_{ES_i} \| \hat{c}_i \| T_{ES_i}))$. Any modification to the message can be checked by verifying the signature. Therefore, the integrity of data transmitted between ES s and CC can be guaranteed by the batch verification mechanism of signatures.

4) Differential Privacy: To resist differential attacks, the proposed scheme implements differential-privacy protection for sensed data. In our scheme, ES_i adds the Laplace noise $\hat{m}_i$ to the aggregated ciphertext. According to the [29, Axiom 2.1.1] (this axiom has been observed to hold in differential privacy), the dataset satisfies the ϵ -differential privacy after encryption. CC finally gets the sum of all S s' data, i.e., $\hat{M}_i = \sum_{j=1}^n m_{ij} + \hat{m}_i$. According to the definition of differential privacy, the randomized algorithm $K : K(D) = f(D) + \text{Lap}(\frac{\Delta f}{\epsilon})$ provides ϵ differential privacy, where D denotes a dataset. The noise $\hat{m}_i$ satisfies the $\text{Lap}(\frac{\Delta f}{\epsilon})$ Laplace distribution. So the decrypted ciphertext $\hat{M}_i$ meets the ϵ -differential privacy under the randomized algorithm K . For the total usage data $M = \sum_{i=1}^l M_i = \sum_{i=1}^l \sum_{j=1}^n m_{ij} + \sum_{i=1}^l \hat{m}_i$, the noise $\hat{m}_i$ is drawn independently from the Laplace distribution. Since the variance is $2(\Delta(f)/\epsilon)^2$, we can know that the variance of $\sum_{i=1}^l \hat{m}_i$ is $2l(\Delta(f)/\epsilon)^2$. We say that the sensitivity of $\sum_{i=1}^l \hat{m}_i$ is $\Delta f'$, where $\Delta f' = l\Delta f$, and the privacy budget is ϵ' . From the equation $2l(\Delta(f)/\epsilon)^2 = 2(\Delta(f)'/\epsilon')^2$, we can know that $\epsilon' = \sqrt{l}\epsilon$, and the total usage data $M = \sum_{i=1}^l M_i = \sum_{i=1}^l \sum_{j=1}^n m_{ij} + \sum_{i=1}^l \hat{m}_i$ satisfies $(\sqrt{l}\epsilon)$ differential privacy. The above analysis can be seen in [30].

5) Resistance to Replay Attacks: To resist replay attacks, PPDA adopts timestamps to ensure the freshness of messages. In the registration phase, ES s and S s apply for registration by sending messages containing timestamps. According to the timestamp, CC can distinguish if the received registration

TABLE III
SECURITY COMPARISON

Attributes	PPDA	[8]	[9]	[26]	[27]	[28]	[11]
Data integrity	✓	✓	✓	✓	✓	✓	✓
Nonrepudiation	✓	×	✓	✓	✓	✓	✓
Privacy protection	✓	✓	✓	✓	✓	✓	✓
Differential privacy	✓	×	✓	×	×	×	×
Identity authentication	✓	✓	✓	✓	✓	×	✓
Efficient batch verification	✓	✓	✓	×	✓	✓	✓
Resistance to replay attacks	✓	✓	✓	✓	✓	✓	✓
Aggregation correctness	✓	✓	✓	✓	✓	✓	✓

messages are legal requests or replay requests. Besides, all ES s can check if the timestamp in the broadcast message is fresh to resist replay attacks during the data collection. Similarly, S s can also resist replay attacks. In the data generation and reporting phase, S_{ij} inserts the timestamp T_{ij} into the signature δ_{ij} and the message to prevent the report data from replay attacks. Finally, all ES s check the freshness of the timestamps in the t received messages to verify that messages are legitimate instead of the replay messages from attackers. In conclusion, our scheme can resist replay attacks.

C. Security Comparison

Based on the above security analysis, PPDA achieves all five security features, i.e., privacy protection, identity authentication, data integrity, differential privacy, and resistance against replay attacks. The security features of our PPDA and the related schemes are summarized and compared in Table III.

In [8], the Paillier cryptosystem and bilinear pairing are used to guarantee the privacy of the individual user and achieve the authentication of messages, respectively. Data integrity and resistance to replay attacks are ensured by using the hash function and timestamp, respectively. However, it cannot achieve differential privacy. The scheme [9] achieves similar security features while using different methods. Specifically, the scheme employs the double trapdoor decryption cryptosystem to achieve the privacy protection of the user and adopts bilinear maps to support authentication and data integrity verification. Regarding replay attacks, the authors also add the time period information into messages to ensure the message's freshness. In [26], Shen et al. use the Paillier cryptosystem to protect the privacy of users. The bilinear pairing and hash function can ensure authentication and data integrity. The uploaded messages contain timestamps, which make the scheme resistant to replay attacks. However, this scheme also cannot achieve differential privacy. In [27], Vahedi et al. adopt an elliptic-curve-based ElGamal cryptosystem to encrypt the message to realize the protection of user privacy. Based on the idea of a digital signature, the scheme adopts bilinear pairing to realize the authentication between communication entities. Data integrity and resistance to replay attacks are achieved by using hash functions and timestamps, respectively. However, the scheme still cannot achieve differential privacy. In [28], the privacy of a user can be protected by the Paillier cryptosystem. Furthermore, the transmitted messages always contain the timeslot information to resist replay attacks. However, the identity cannot be verified by the receiver and the differential attacks cannot be prevented. In [11], the Paillier cryptosystem

is also used to protect the privacy of a single user. Additionally, the authors utilize a modified BLS signature to achieve data integrity and identity authentication. Moreover, they also insert the timestamp into a message transmitted between communication entities to ensure the freshness of the message, so as to resist the replay attacks.

VI. PERFORMANCE EVALUATION

To evaluate our scheme, we illustrate the comparison of our scheme with six other related schemes [8], [9], [11], [26], [27], [28] in terms of communication overhead and computational cost. For the sake of simplicity, we denote our scheme as PPDA in the following description. Besides, in order to better describe the communication and computational cost of the compared schemes, we use ES_i and CC to replace the edge devices and computing center in schemes [8], [9], [11], [26], [27], [28], respectively.

A. Experimental Setting

In the experiments, we set the number of CC , ES , and S are 1, 2, and 20, respectively. Moreover, the number of subordinate S s for each ES is set to 10. We compare some operations including Encryption (Enc), Decryption (Dec), Signature (Sig), Single verification (S-Verif), Batch verification (B-Verif), Bilinear mapping (Bili-map), and Aggregation (Agg).

For fair comparisons between PPDA and six other related schemes, the detailed parameters are set fairly as follows. The length of big prime numbers p_0, q_0 in the Paillier cryptosystem are both set to 1024 bits, i.e., $k_1 = 1024$. The length of an element in $G_q(a, b)$ of the ECDSA signature is assumed to be 256 bits, i.e., $k_2 = 256$. With respect to the identity and timestamp, the length is set to 2 bytes and 4 bytes, respectively. Therefore, the ciphertext size of Paillier and the hash function H_1 are 256 and 32 bytes, respectively. Additionally, the length of feedback from CC to ES s and S s in [9] is set to 4 bytes. In bilinear mapping, the length of an element in the cycle group is 128 bytes.

Based on the above setting, we conduct extensive experiments on a laptop with Windows 10 Professional operating system, a 2.80 GHz Intel(R) Core(TM) i7-7700 CPU, and 16-GB RAM. The time cost for the bilinear pairing operation is simulated based on Java with the JPBC library.

B. Communication Overhead Comparison

In this part, we compare the communication overhead of PPDA with other related schemes [8], [9], [11], [26], [27], [28].

1) *Communication Overhead Between S_{ij} and ES_i* : In PPDA, S_{ij} receives the message $\{ID_{CC}, ID_{ES_i}, T_{QES_i}, \beta, \delta_{QES_i}\}$ from ES_i in the data collection request and sends the message $\{ID_{S_{ij}}, \delta_{ij}, T_{ij}, c_{ij}\}$ to ES_i during the data generation and report. Therefore, the communication overhead between S_{ij} and ES_i is totally 644 bytes. While in [9], U_i sends $(t, U_i, C_{i,t}, \delta_{i,t})$ to the fog F_k and receives $(t, fb_k, \delta_{k,t}^{fb})$. As a result, the communication overhead is 1294 bytes in total. In [8], [11], [26],

TABLE IV
COMMUNICATION OVERHEAD COMPARISON (BYTES)

Attributes	PPDA	[8]	[9]	[26]	[27]	[28]	[11]
S-ES	644	582	1294	650	390	674	580
ES-CC	604	582	1704	671	390	704	644
Total overhead	1248	1164	2998	1321	780	1378	1224

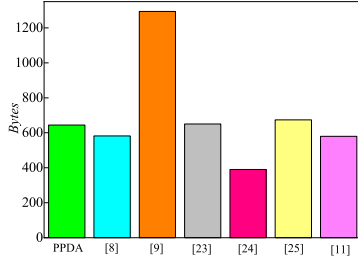


Fig. 3. Communication overhead between S and ES .

[27], and [28], S_{ij} sends $\{c_i, V_i, t_i, id_i\}$, $\{CT_i, \sigma_i, T\}$, $\{C_{ik} || ID_{RAGW_i} || ID_{User_{ik}} || T || \delta_{ik}\}$, $\{R_{it} || S_{it} || t || ID_{U_i} || \delta_{it}\}$, and $\{C_i || M_i^{gw} || M_i^{cs} || N_i || X_i || ID_i\}$ to ES_i , respectively. Therefore, the corresponding communication overhead between S_{ij} and ES_i of these schemes are 582, 580, 650, 390, and 674 bytes, respectively.

The comparison of communication overhead between ES_i and S_{ij} is illustrated in Table IV and Fig. 3. Based on the figure, the communication cost in our scheme is evidently lower than that in [9], [26], and [28] and is close to that in [8] and [11] but is higher than that in [27]. Specifically, our work can resist differential attacks, while [8] cannot achieve this.

2) **Communication Overhead Between ES_i and CC:** In our scheme, ES_i receives the message $\{ID_{CC}, T_{QCC}, \delta_{QCC}, \beta\}$ from CC and sends $\{ID_{ES_i}, \delta_{ES_i}, T_{ES_i}, \hat{c}_i\}$ to CC. So the communication cost between ES_i and CC is 604 bytes. While in [9], fog F_k sends messages $(t, F_k, AD, c_{k,t}, \delta_{k,t})$ and $(\Delta, t, F_k, c_{k,t}, \delta_{k,t})$. Simultaneously, it also receives the message $(t, fb_k, \delta_{k,t}^{fb})$. Therefore, the communication overhead is 1704 bytes. Besides, in [8], [11], [26], [27], and [28], S_{ij} sends $\{c_i, V_i, t_i, id_i\}$, $\{CT, \sigma, \xi, T\}$, $\{SAMDD || ID_{CC} || ID_{DGW} || n_1 || n_2 || \dots || n_m || T || \delta\}$, $\{R_t || S_t || t || ID_{AG_i} || \delta_{AG_i,t}\}$, and $\{C_T || M_T^{CS} || X_T || N_T\}$ to CC. Therefore, the communication overhead between ES_i and CC of these schemes are 582, 644, 671, 390, and 704 bytes, respectively.

Table IV and Fig. 4 demonstrate the comparison of communication between ES_i and CC. It is clear that our scheme PPDA is more efficient than that in [9], [11], [26], and [28] and is close to that in [8] while the communication overhead of PPDA is higher than that in [27]. The main reason is that our PPDA allows the sensors to upload the data reports only when receiving the data upload request instead of uploading data reports automatically. To ensure the security of the data upload request, more messages should be sent, i.e. the signature of the request. Even though heavier communication overhead is caused in this way, it is more flexible and energy efficient.

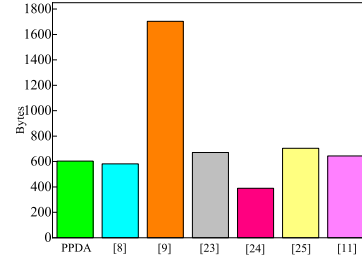


Fig. 4. Communication overhead between ES and CC .

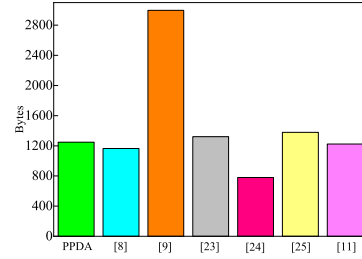


Fig. 5. Communication overhead in total.

TABLE V
TIME COST ON S_{ij} (MS)

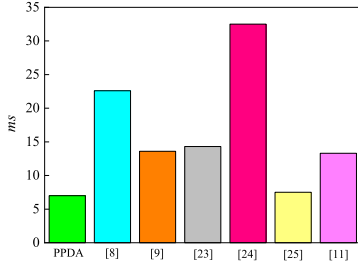
Attributes	PPDA	[8]	[9]	[26]	[27]	[28]	[11]
Enc	2	17.3	5.8	3.3	18.5	4.9	4.1
Dec	-	-	-	-	-	-	-
Sig	3	1.3	7.8	11.0	14.0	2.6	9.2
S-Verif	2	-	-	-	-	-	-
B-Verif	-	-	-	-	-	-	-
Bili-map	-	4	-	-	-	-	-
Agg	-	-	-	-	-	-	-
Total time	7	22.6	13.6	14.3	32.5	7.5	13.3

3) **Total Communication:** The total communication overhead between S and ES , and between ES and CC is compared in Table IV and Fig. 5. From the figure, we can see that PPDA outperforms schemes [9], [26], [28] and is almost similarly efficient to schemes [8], [11] regarding the communication overhead but is less efficient than [8]. However, Chen et al. [8] achieves fewer security features than our PPDA and suffer more computational costs.

C. Computational Cost

In this part, we compare PPDA with other related schemes [8], [9], [11], [26], [27], [28] on the computational cost by conducting extensive experiments.

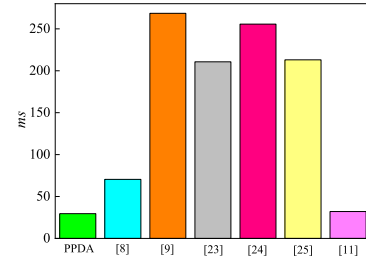
1) **Time Cost at S_{ij} :** We summarize the computation cost of a single S_{ij} of PPDA and schemes [8], [9], [11], [26], [27], [28] in Table V. For better observation, we also show the time cost of S_{ij} in Fig. 6. From the table and the figure, we can see that the computational cost of S_{ij} in PPDA is 6.7% less than that of related schemes on average. There are different

Fig. 6. Computational cost at S_{ij} .TABLE VI
TIME COST ON ES_i (MS)

Attributes	PPDA	[8]	[9]	[26]	[27]	[28]	[11]
Enc	-	-	-	-	0.5	-	-
Dec	-	-	-	-	-	4	-
Sig	3	28.1	122	30.1	30.2	205	-
S-verif	2	28.2	-	-	-	-	-
B-verif	24	-	132.8	180.4	215.1	-	12
Bili-map	-	-	13.7	-	-	-	-
Agg	0.5	1.4	0.1	0.2	9.8	4	20
Total-time	29.5	70.4	268.6	210.7	255.6	213	32

reasons in different related schemes for this result. In [8], a sensor S_{ij} similarly performs the encryption and signature. However, it needs to calculate multiple time-consuming large-number multiplications and bilinear mappings to encrypt multiple types of data and construct signatures, respectively. Although Liu et al. [9] avoids the use of bilinear pairing in S_{ij} , it adopts a more complex double trapdoor decryption system, which causes more computation resources than the Paillier cryptosystem. In [26], S_{ij} similarly performs the encryption and signature, but it additionally needs to pack the multitype data before encryption. In [11], S_{ij} also needs to pack the multitype data into a plaintext like [26], which enlarges the plaintext to be encrypted and brings additional overhead. Moreover, the signature process involves two power exponential operations, which are the main part of the computation cost of S_{ij} . In [27], the sensed data need to be mapped into a point of an elliptic curve before being encrypted with the ElGamal cryptosystem. However, the mapping takes more computational time as it is realized by point multiplication defined on an elliptic curve. In [28], the constructions of the authentication information and fault tolerance guarantee are really time-consuming.

2) **Time Cost at ES_i :** We sum up the computational cost of ES_i in PPDA and other related schemes [8], [9], [11], [26], [27], [28] in Table VI. From this, we can find that the computation of ES_i in PPDA is more efficient than all the related work. Specifically, the computational cost has been reduced by 16.9% on average. It is due to the fact that in PPDA, we adopt a batch signature verification algorithm to speed up the verification process while the most time-consuming operations are multiplications on large integers, i.e., the aggregation of the Paillier ciphertexts. Therefore, few time-consuming operations are performed in PPDA. However, in other schemes e.g., [8], [9], [11], [26], [27], some operations with heavy computation are needed to aggregate the signatures, such as multiple power

Fig. 7. Computational cost at ES_i .TABLE VII
TIME COST ON CC(MS)

Attributes	PPDA	[8]	[9]	[26]	[27]	[28]	[11]
Enc	-	-	-	-	0.5	47	-
Dec	3	31.8	89.0	0.8	49.4	2	46
Sig	28	-	-	-	-	-	-
S-verif	24	29.4	89.4	34.3	-	2	45
B-verif	-	-	-	142.3	135.9	-	-
Bili-map	-	13.7	-	-	-	-	-
Agg	-	-	-	0.2	-	-	-
Total time	55	74.9	178.4	177.6	185.8	51	101

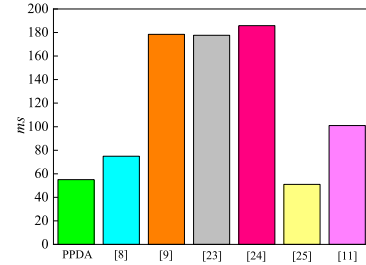


Fig. 8. Computational cost at CC.

exponential and bilinear mapping operations. For a more intuitive comparison, we describe the time cost of ES_i in Fig. 7.

3) **Time Cost at CC:** We generalize the computational cost of CC in PPDA and schemes [8], [9], [11], [26], [27], [28] in Table VII. According to the table, our PPDA has improved 26.8% of the computational cost of CC compared to that in [8], [9], [11], [26], and [27]. Although CC in all the schemes performs both signature verification and decrypts the aggregated ciphertexts, CC in PPDA conducts batch signature verification not by bilinear pairing operations but only by point addition and three times point multiplications while in other schemes, CC needs to perform more time-consuming bilinear pairing or point multiplication operations for signature verification. The corresponding computational cost in [28] is a little better than that of PPDA. That is because, in PPDA, we strengthen the security features at the cost of increasing the computational cost, i.e., differential privacy and identity authentication. As a result, our scheme achieves stronger security than that in [28]. For better observation, we describe the time cost in CC in Fig. 8.

VII. CONCLUSION AND FUTURE WORK

In this article, we propose an efficient and robust privacy-preserving data aggregation scheme for edge-supported IIoT, namely PPDA. In particular, PPDA adopts the Paillier cryptosystem to protect the data privacy of S s and utilizes differential privacy to resist differential attacks. Moreover, PPDA not only ensures the message integrity and identity authentication of S s, ES s, and CC by employing the modified ECDSA signature but also enables ES s and CC to efficiently verify the signatures via the batch signature verification algorithm. Finally, we analyze the security features of our PPDA, and the result demonstrates that PPDA meets the five security requirements described before. Extensive experiments are conducted to evaluate PPDA, and the results illustrate that PPDA is efficient in both communication and computation. Specifically, compared to the related schemes, PPDA has improved 6.7%, 16.9%, and 26.8% of the computation efficiency on the S , ES , and CC sides on average, respectively. Terminal devices in the smart grid usually have very limited computational and communication resources. However, the existing work is far from sufficiently lightweight for practicality. In future work, we will further study lightweight data aggregation schemes to reduce the computation and communication overhead on both the ES and the S sides for more practicality.

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